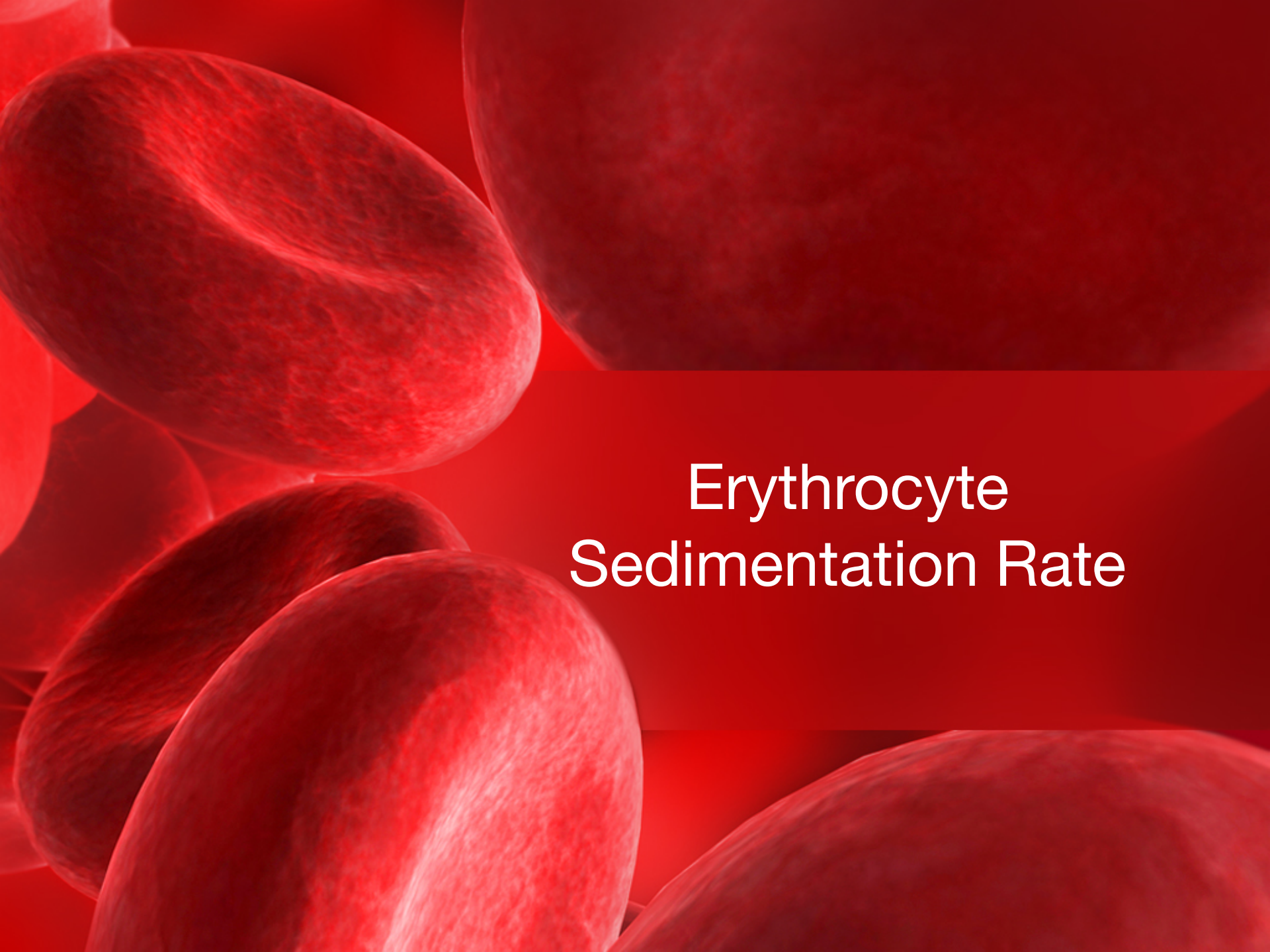
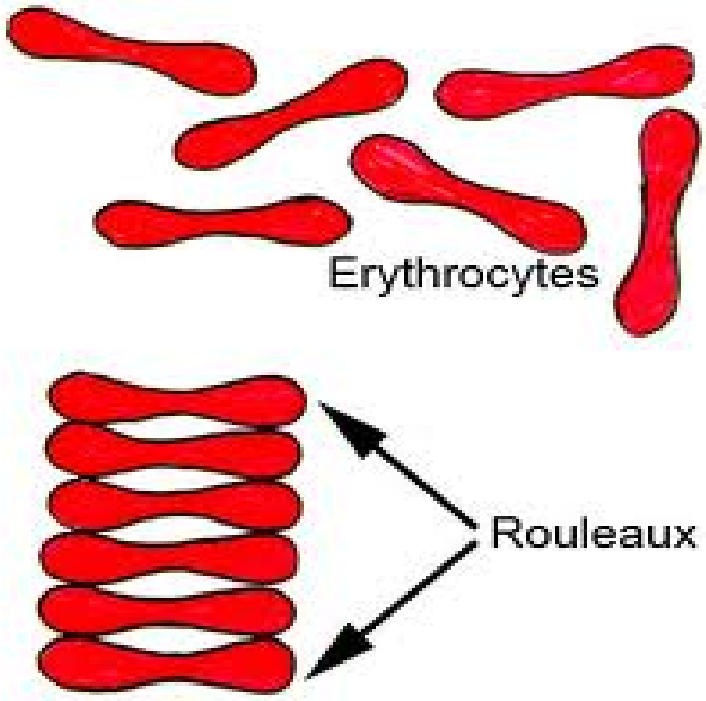

Erythrocyte
Sedimentation Rate

Erythrocyte Sedimentation Rate

Definition:

It is the rate of sedimentation of RBCs in anticoagulated blood kept in a vertical narrow tube. RBCs sedimented to the bottom of the tube, leaving a layer of clear plasma above it.

The RBCs sediment because their density is greater than that of plasma; this is particularly so, when there is an alteration in the distribution of negative charges on the surface of the RBC (which normally keeps them separate form each other) resulting in their gathering to form large aggregates known as rouleaux



Determinants of ESR:

1) Plasma proteins:

↑ globulins or fibrinogen…helps rouleaux formation .. increase ESR

↓ globulins or fibrinogen….. decrease ESR

↓ albumin….. helps rouleaux formation…

increase ESR

2)Viscosity:

Increased viscosity retards ESR, whereas decreased viscosity enhances it.

↑ Temperature → ↓ viscosity→ ↑ ESR and vice versa.

3) RBCS count:

* Anemia …….fast sedimentation… ESR

* Polycythemia .....slow sedimentation.. ESR

4) Shape of the RBCs:

- Normal biconcave shape of RBCs facilitates rouleaux formation.

- **In sickle cell anemia and in spherocytosis**, the shape of the cells tends to prevent rouleaux formation and hence decreases ESR.

- **In iron deficiency anemia,** the microcytes resist rouleaux formation and sedimentation is retarded.

N.B: The shape of RBCs in iron deficiency anemia, anemia due to spherocytosis, and sickle cell anemia, reduces rouleaux formation. Therefore, the ability of RBCs to sediment is reduced which compensate for the decreased blood viscosity and usually the

Determination of ESR:

Westergren's method

Requiremets:

1. Westergren's tube:Graduated 0-200 mm from above downwards.
2. Stand for westergren's tube.
3. Timer or watch.



Procedure:

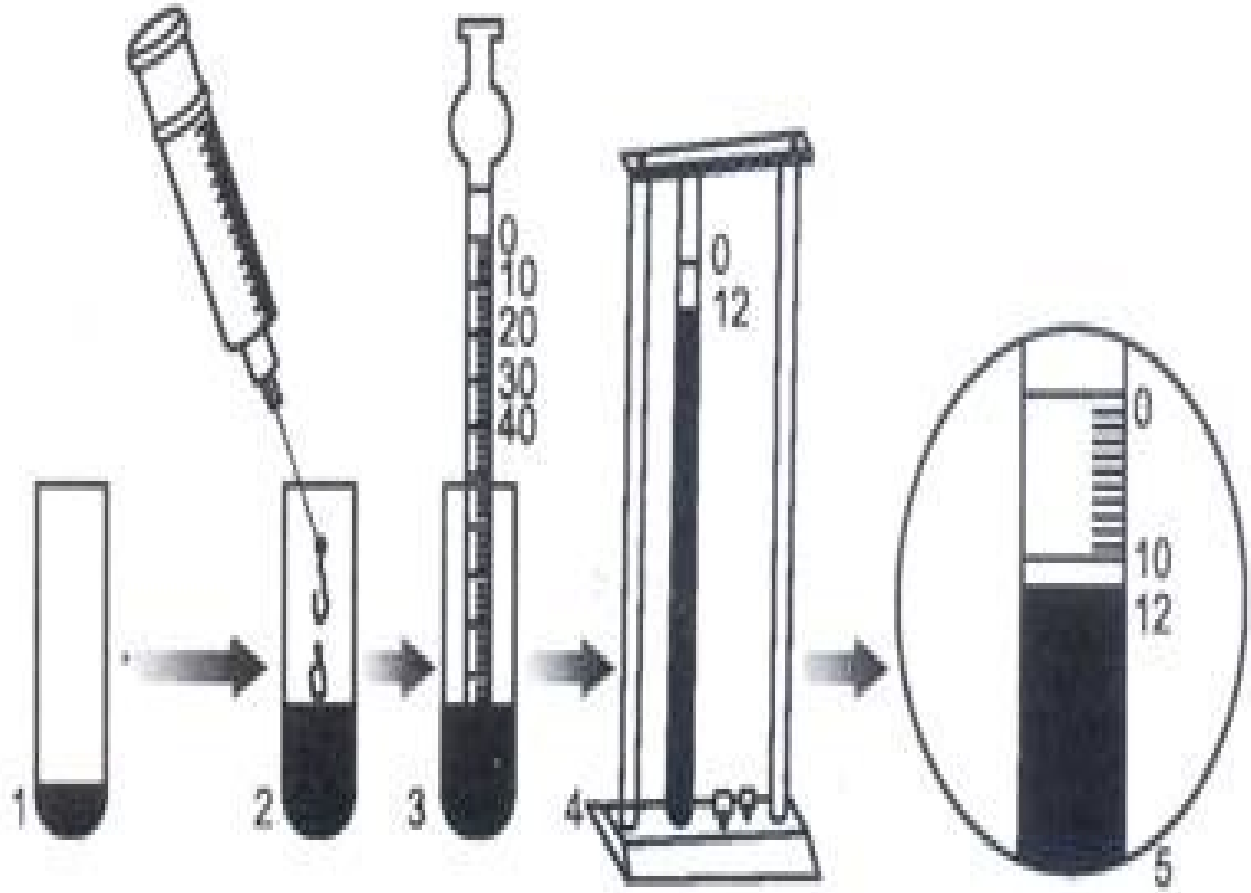
1. Mix 2 ml of venous blood to 0.5 ml of 3.8% sodium citrate solution. This will give a blood : citrate ratio of 4 : 1

2. Fill the Westergren's pipette with blood - citrate mixture by sucking till bringing the blood column to exact zero mark.

3. Transfer it to the Westergren's stand and press its lower end firmly into the rubber cushion, then slit the upper end of the pipette under the screw cap. Confirm that there's no leakage of blood.

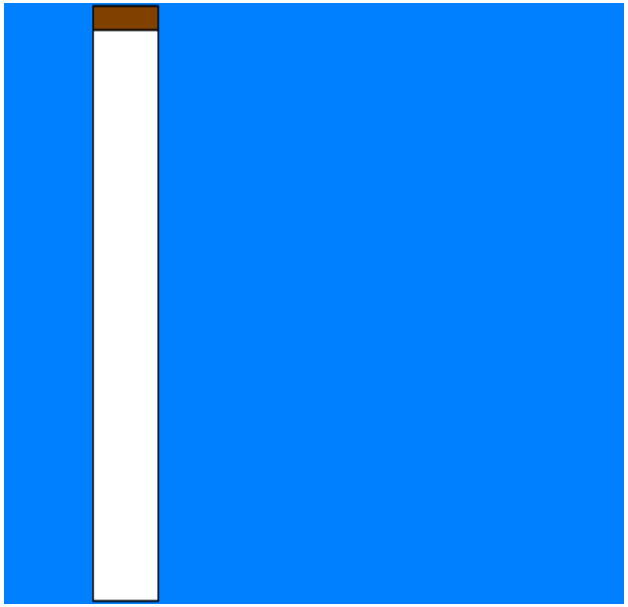
4. Leave the pipette for one hour & then calculate the ESR by measuring the distance between the top of the plasma and the top of the sedimented red blood cells.

Express your results as: …….. mm/h.


0
10
20
30
40
0
12
0
10
12
1
2
3
4
5

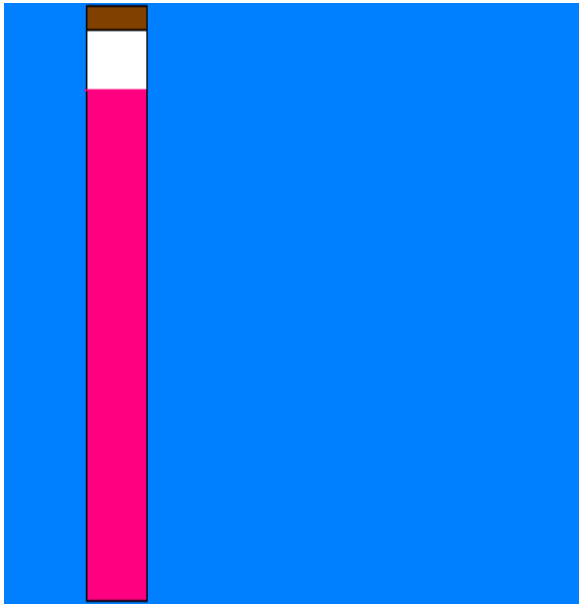
How is the test performed?

Take a tube

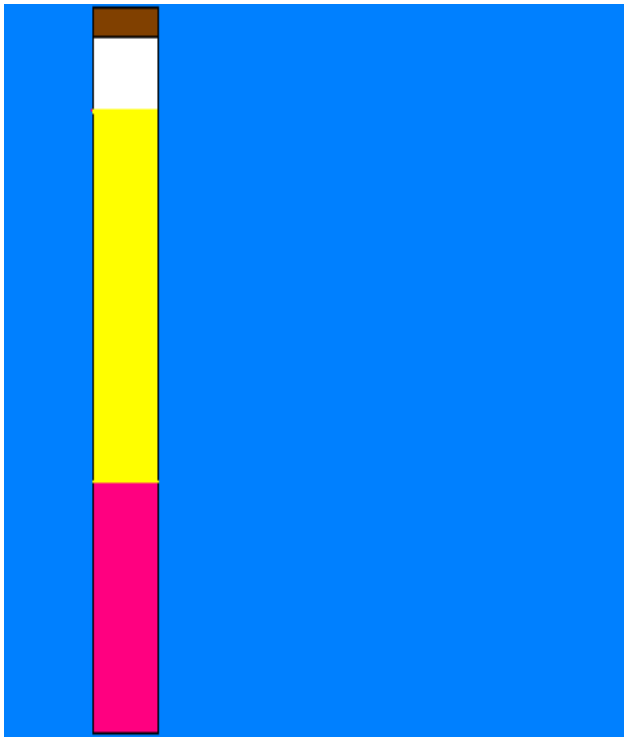


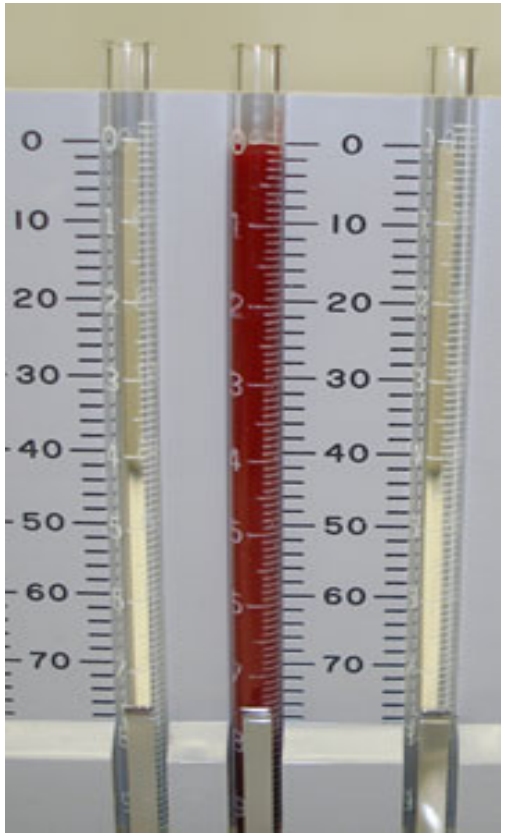
Fill with blood

Leave it for an hour

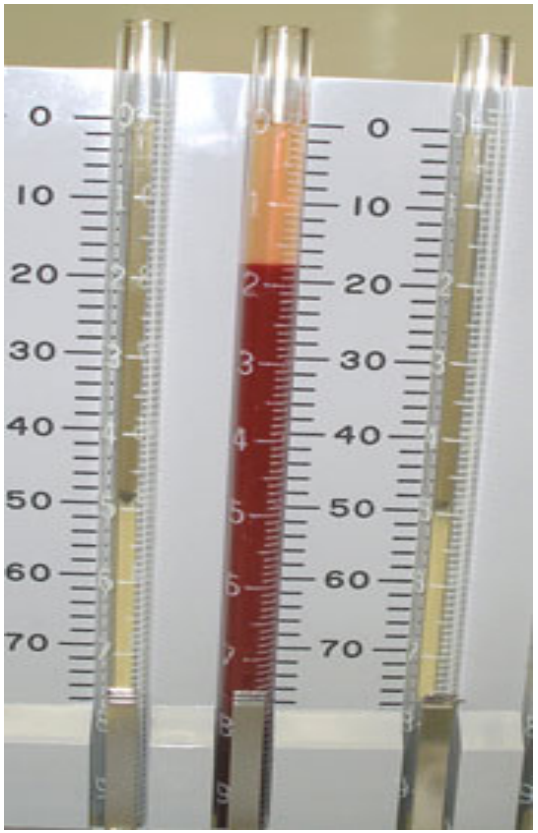


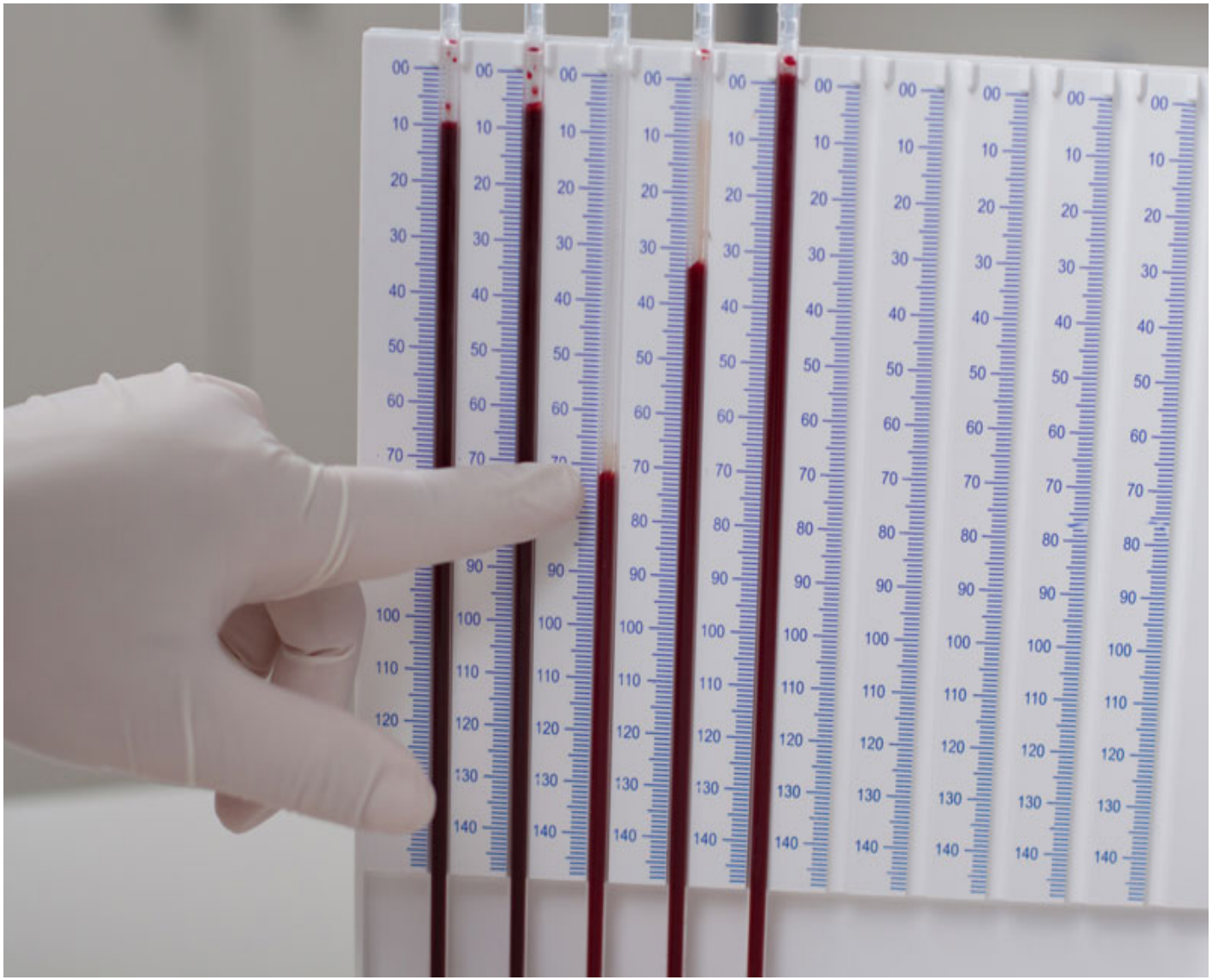
Sedimentation occurs





1 hour





Normal values:

- ✔ Males: 4-6 mm/h
- ✔ Female : 8-10 mm/h.

- N.B: ESR can be calculated after 2 hours using the following equation:

$$\frac{\text{Reading of 1st hour} + \frac{1}{2}\text{ reading of 2nd hour}}{2}$$

Factors affecting the ESR:

Increase ESR		Decrease ESR
Physiological	**Pathological**	Low protein: fibrinogen or gamma-globulins
Female during Pregnancy , Menstruation & Lactation	Infection, Inflammation, Malignancy, Fractures …. (elevated gamma-globulins or fibrinogen)	Polycythaemia
After exercise	Anemia	Spherocytosis, Sickle cell (inhibit rouleaux formation)

Clinical significance of ESR:

- Many factors affect the ESR making it extremely non-specific.

- So, ESR is **not Diagnostic**

but it is **Prognostic**

- That helps in monitoring disease activity or response to therapy


Osmotic Fragility

:Definition

RBCs fragility is the lack of resistance of RBCs to hemolysis when exposed to hypotonic saline solutions (osmotic fragility) or when subjected to mechanical trauma (mechanical fragility).

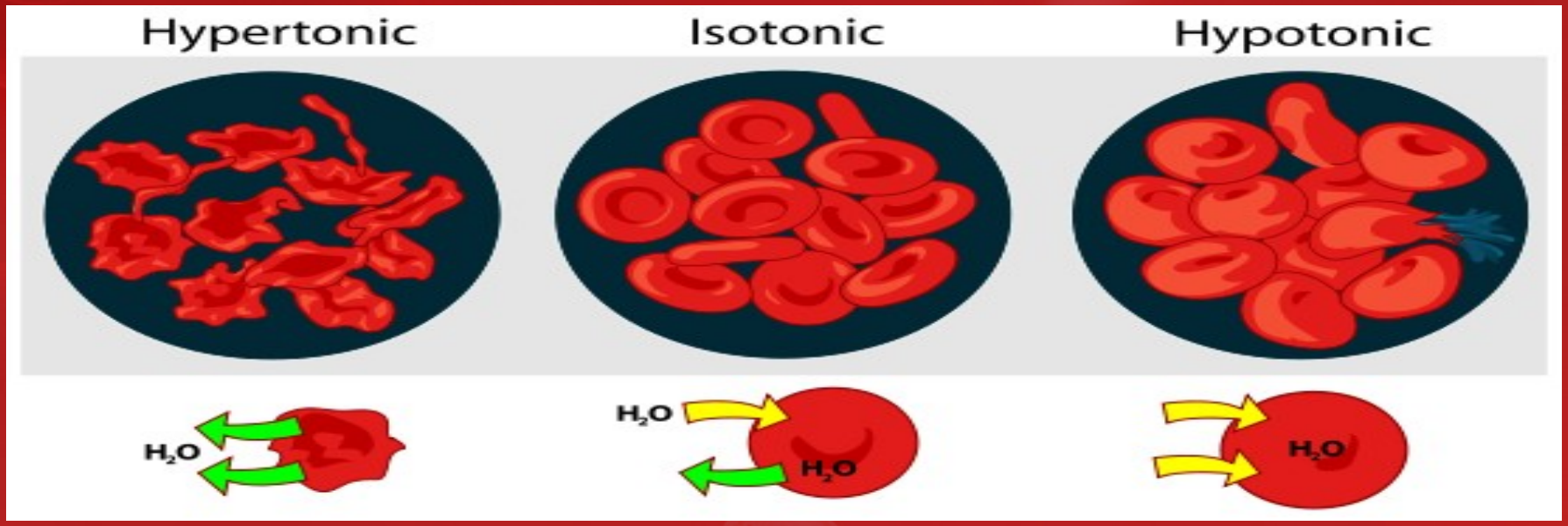
Normally, the RBC has high surface area to volume ratio (due to its biconcave shape) which allows it to increase in volume before being rupture when placed in hypotonic saline, but spherical RBCs hemolyze easily because of low surface area to volume ratio.

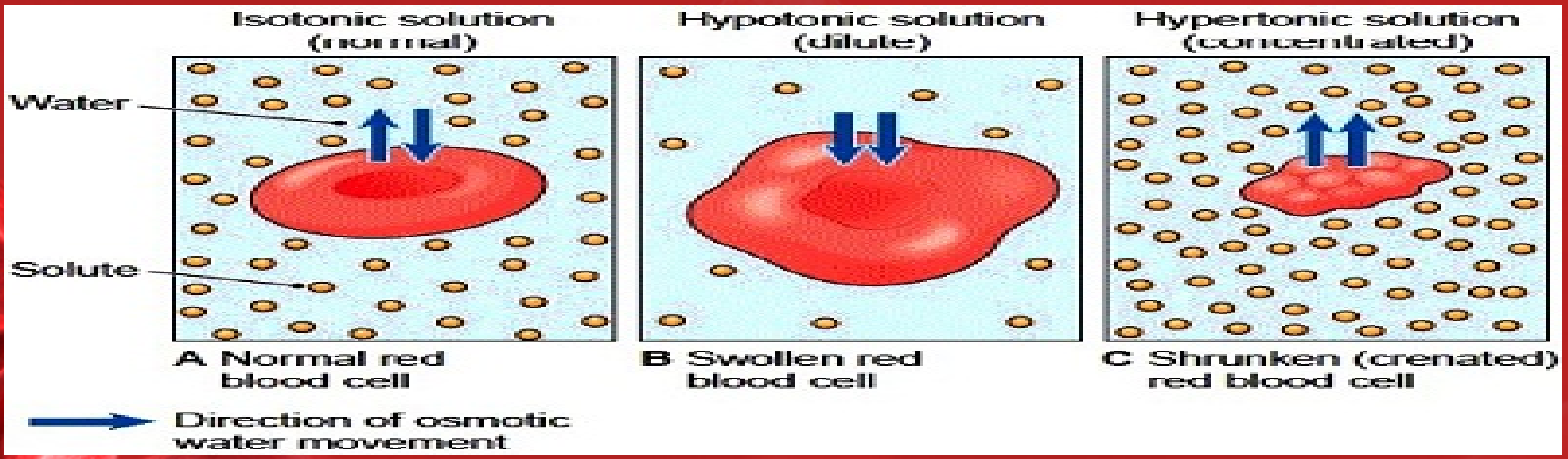
Osmotic fragility: is a test to detect red blood cells that are more fragile than normal.

It is determined by placing RBCs in hypotonic NaCl solutions with different levels of dilution and measuring the amount of hemolysis.

Osmotic fragility

EFFECT OF HYPERTONIC, ISOTONIC AND HYPOTONIC SOLUTIONS ON RED BLOOD CELLS:





Determination of osmotic fragility

Method

1. Prepare a series of hypotonic solutions with NaCl content ranging from distilled water to 0.8 %, to which a small amount of fresh heparinized blood is added.

2. After 30 min. of incubation followed by centrifugation, the saline concentrations at which start of hemolysis and complete hemolysis occur are determined.

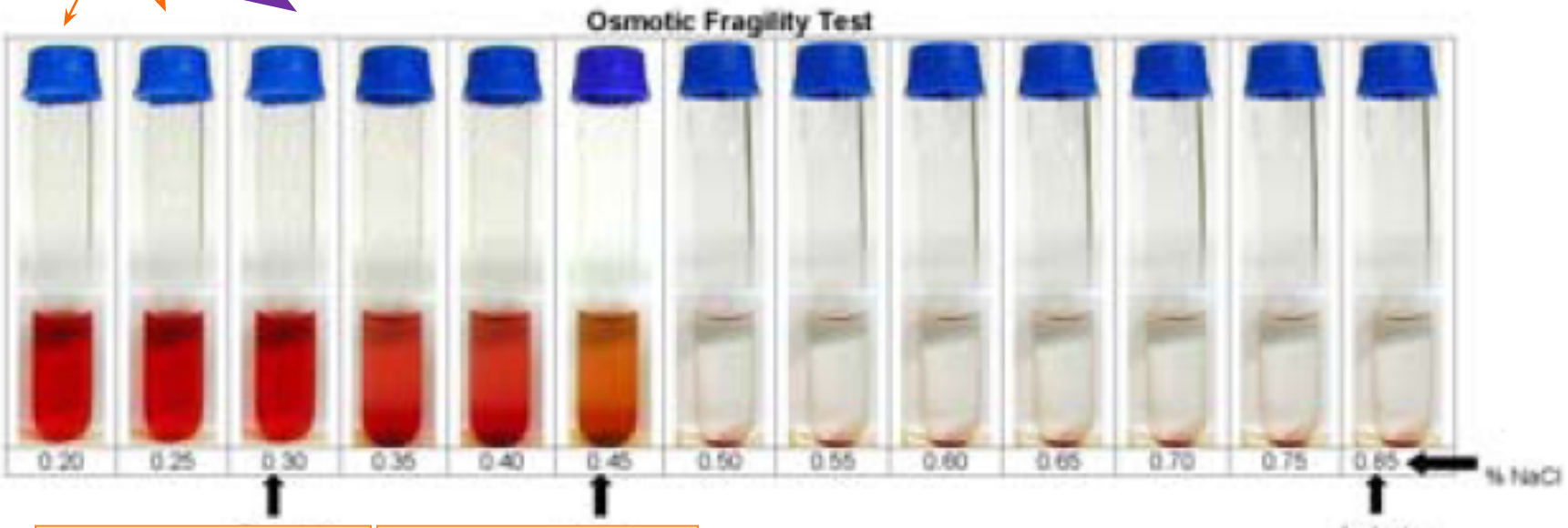
3. Start of hemolysis is determined by recording the pink coloration (hemoglobin color) of the supernatant.

4. As the osmotic lysis progressively increases, the supernatant color gradually darken from pink to red till complete hemolysis is reached and the red color of the supernatant is fixed.

HOW TEST IS PERFORMED?

Reddish coloration of sol. + no ppt.

Reddish coloration of sol. + ppt.

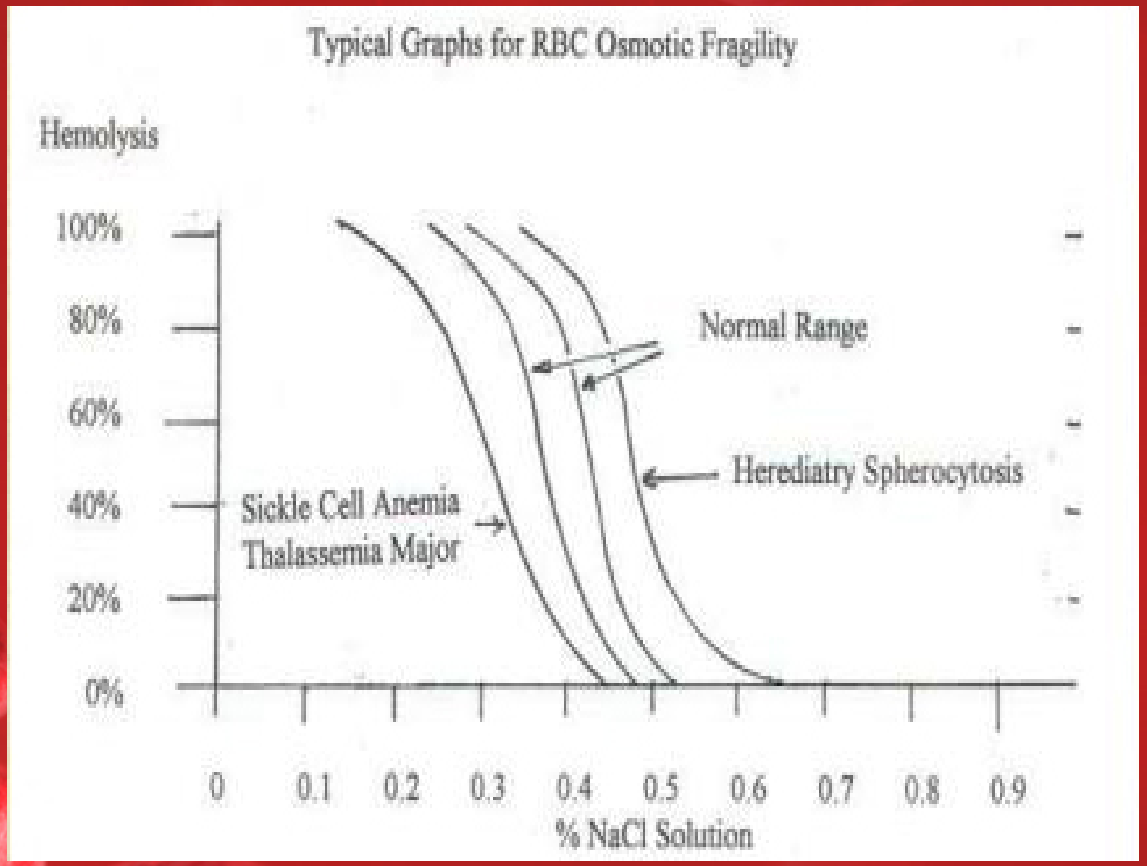


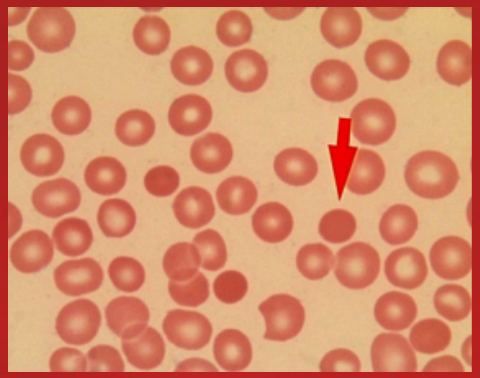
NORMAL VALUES:

- Normally red blood cells begin to hemolyze at 0.48% saline, 50% lysis occur at 0.4-0.42 and lysis completed at 0.35% saline.

SIGNIFICANCE OF THE TEST:

- To detect <u>hereditary spherocytosis</u>
- (hemolysis starts at 0.7% and completed at 0.5% saline solutions)



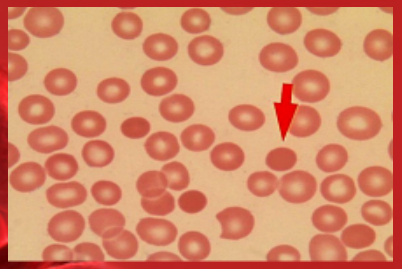






CAUSES OF RED BLOOD CELLS HEMOLYSIS:

- corpuscular
  - Defect in cell wall (congenital spherocytosis)
  - Defect in Hb (sickle cell anemia)
- Extra corpuscular
  - drugs
  - Chemical poisons
  - infections
  - Antibodies against Rbcs as incompatible blood transfusion)





Thank You